

Problem 1: DNN and Wine Classification (80 points)

(e) [Pro/Dis] (10 pts) Can you further reduce model size beyond parts (b), (c), and (d)? If yes, implement your proposed method and compare results. If not, explain why.

Yes we can!

My proposed idea was to combine all three steps together in an unprecedented technique that has never been seen before as long as you haven't done lab 3

We take the student model that created in part D, and then run it through the other techniques to reduce its size. First we prune, then we quantize. (because you can't prune a TFLite model) (also it makes more sense in general to cut up the model architecture and then compress it) (none of these are done to the teacher because we want that model to keep as much complexity as possible) We also quantize to Int 8 specifically since it's by far the most dramatic reduction in size

Model	Accuracy	Size
Part B: INT 8	1.00	5.70 KB
Part B: FLOAT16	1.00	14.03 KB
Part B: DYNAMIC	1.00	14.03 KB
Part C: Stripped Sparse	1.00	8.140625 KB
Part D: Student Model	1.00	6.0390625 KB
Part E: Stripped Student	1.00	6.04296875 KB
Part E: Stripped INT8 Student	1.00	3.56 KB

Problem 2: Exploring Edge Impulse (20 points)

1. [Dis] (5 pts) What types of "Learning Blocks" are available under the free-tier Edge Impulse account (e.g., supervised learning—classification and regression, unsupervised learning—clustering, self supervised learning, etc.)?

For each type, provide at least one example model that can be implemented using the free tier.

- Classification
 - Classifying movement type (walking, running, jumping) based on accelerator data
- Regression
 - Predict the temperature tomorrow based on past data
- Anomaly Detection
 - Detect if a piece of factory equipment is showing signs of deviant behavior (aggressive/unusual vibration, atypical temperature spikes)

2. [Dis] (5 pts) List all the types of layers that can be used under the “Classifier” section in the free-tier Edge Impulse. Briefly explain the purpose of each layer identified.

- Dense
 - Your standard fully connected neural network layer. Used for learning patterns
- 1D Convolution / Pooling
 - Feature extraction from sequential data
- 2D Convolution / Pooling
 - Feature extraction from 2D data (images)
- Reshape
 - Change the shape of your layer into one compatible with the next guy in the pipeline
- Flatten
 - Collapse data into a 1D array
- Dropout
 - Not really a layer. A technique done while training that randomly zeroes out neurons to cut em down. Prevents overfitting

3. [Dis] (5 pts) What types of model compression options are available under the “Deployment” section in the free-tier Edge Impulse? List all available methods.

- Seems like the only compression options are int8 and float32. However, you can change the inference engine. I guess.

4. [Dis] (5 pts) What input modalities are supported for synthetic data generation in the free-tier version of Edge Impulse? For each modality, list the synthetic data generation methods/algorithms provided. Briefly explain why synthetic data is important in machine learning applications

- Its an alright way to start or pad out a dataset that is unable to be collected the natural way (either due to difficulty or time limitations)
- Image Data
 - Dall-e
 - RealVisXL
- Audio
 - Google TTS
- Time-Series
 - MATLAB

- PyBullet

There are also these “public blocks” which seem different than the above but aren't listed in the documentation website

PUBLIC BLOCKS

Edge Impulse Inc. / Black Forest Labs Flux-Pro Synthetic Image Generator

Uses Replicate.com's API for the Flux Pro model from Black Forest Labs to generate "State-of-the-art image generation with top of the li...

Edge Impulse Inc. / DALL-E 3 Synthetic Image Generator

Use OpenAI's DALL-E 3 to create synthetic images for your project.

Edge Impulse Inc. / Whisper Synthetic Voice Generator

Uses OpenAI's text-to-speech model to generate synthetic voice samples. Very useful for building keyword spotting models for arbitrary...
